

PROJECT		JOB NO.				
CONTRACTOR :						
SUB-CONTRACTOR : Winner Engineering Pte Ltd						
TRADE	: ACMV SERVICES		DRAWING NO :			
FUNCTIONAL- TESTING COMMISSIONING CHECKLIST (FAN COIL UNIT)						
Unit Designation :						
Unit Make :						
Model no.						
Location :						
Serial no :						
S/N	Section	Inspection / Test item	CHECKED CONDITION (Please tick inside checkbox)			Remarks
			Yes	No	NA	
1	General	Check that the equipment is clean, free of debris, and undamaged before commencing operations.				
2	Documents	Verify chiller tag, make, model, serial number, refrigerant and rated capacity.				
3	Electrical	Measure incoming supply and verify outgoing motor supply on run command.				
4	Fan & motor	Verify fan start/stop and rotation direction.				
5	Fan & motor	Test low, medium and high fan speeds where provided.				
6	Fan & motor	Test EC motor or VSD speed control where fitted.				
7	Fan & motor	Record motor current on each applicable phase at design duty.				
8	Fan & motor	Check fan/motor bearings, noise, vibration and mounting during operation.				
9	Air leakage	Check FCU casing, return box and duct joints for air leakage at high speed where available.				
10	Controls	Test thermostat display, setpoint adjustment and temperature response.				
11	Controls	Verify control valve full travel and response to cooling demand.				
12	Controls	Verify off-coil, room temperature and RH control functions where specified.				
13	Controls	Test Local/Manual and Remote/BMS operation where provided.				
14	Controls	Verify LCP/LMCP Start, Stop, Run, Trip and lamp-test functions where provided.				
15	BMS	Compare BMS temperature/RH readings with calibrated field measurements where monitored.				
16	BMS	Verify BMS start/stop, status, fault and valve/damper feedback where connected.				
17	BMS	Verify schedules and operating setpoints where controlled by BMS.				
18	Dampers	Test motorized dampers and non-return dampers where fitted.				
19	Dampers	Verify associated VAV/terminal operation and final balancing positions where fitted.				
20	Interlocks	Test applicable fire/smoke shutdown and associated damper interlocks.				

21	Drainage	Test condensate pump and high-level switch/overflow response where fitted.				
22	Drainage	Check condensate drainage with the FCU operating.				
23	Water side	Check coil, pipework and valve joints during water circulation.				
24	Water side	Record CHW supply/return temperatures and water flow or pressure readings as specified.				
25	Airflow	Measure total FCU/main-duct airflow where a suitable measurement point is provided.				
26	Airflow	Measure and balance each supply diffuser/grille at the design fan speed.				
27	Airflow	Measure fresh-air supply to the FCU/served area where applicable.				
28	Cooling	Record return-air, off-coil/supply-air and room temperature; record room RH where required.				
29	Cooling	Check external sweating, cooling stability and noise during the specified test period.				
30	Completion	Record final fan speed, valve/damper positions, setpoints and control modes.				

Comments to follow up Action :

Yes = satisfactory; No = rectify and re-inspect; NA = not applicable (state reason). Blank = not checked.

CHECKED BY:		WITNESSED BY	INSPECTED BY
NAME			
DESIGNATION :			
COMPANY :			
SIGNATURE :			
DATE :			